

Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with 3+ years of experience building and maintaining production applications across AI, CRM, trading, and cloud contact center platforms. I work end-to-end — designing REST APIs, modeling data in MongoDB and PostgreSQL, building React.js/Next.js frontends, and implementing authentication and RBAC/ACL authorization. I've built and integrated a RAG-based chatbot into a live trading platform for contextual financial information retrieval, and most recently led the migration of a legacy Perl-based call-recording workflow to a Python service deployed across 40 production clusters, with staged rollout, feature flags, and rollback safety. Comfortable across Python, FastAPI, Django, Flask, JavaScript/TypeScript, and AWS, with a strong focus on production reliability, troubleshooting, and observability using Datadog and Sentry.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: FastAPI, Django, Flask, REST APIs, API Development, Backend Automation

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2, Microservices

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: MongoDB, PostgreSQL, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

WORK EXPERIENCE

Software Engineer • [CA One Tech Cloud Inc.](#)

Apr 2026 – Present | Bengaluru, India

Client: [Convoso](#) — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services for a high-volume cloud contact center platform, supporting call-recording ETL, data processing, storage workflows, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, supporting feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Built AWS S3 audit and recovery workflows for recording storage, including deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Implemented production observability using Datadog, StatsD, dashboards, and Sentry for monitoring service health, tracking errors, diagnosing processing failures, and supporting production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing troubleshooting and implementing fixes to maintain reliable recording-processing workflows.

Software Engineer • [Mantra Technologies](#)

Dec 2023 – Mar 2026 | Hyderabad, India

- Delivered new features and enhancements across production applications, translating product requirements into application modules, business logic, and end-to-end workflows.
- Developed and integrated end-to-end application functionality across frontend, backend, APIs, and data flows, ensuring features worked consistently across connected application components.
- Implemented authentication, authorization, and access-control workflows, managing permissions across users, roles, and application resources.
- Integrated third-party services and APIs, handling request and response flows, data validation, error scenarios, and integration-related issues.
- Supported releases and production operations through deployment validation, testing, debugging, root-cause analysis, defect resolution, and troubleshooting of application issues.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.

- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated the RAG chatbot into application workflows, enabling conversational access to relevant financial information.
- Built FastAPI/Python backend APIs and business logic for financial data processing and application workflows.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM

- Developed backend modules and REST APIs for an AI-driven CRM using Python and FastAPI, implementing business logic and workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built the Contacts management experience with inline editing, global search across CRM modules, access to related Tasks, Deals, and Documents, and Google Sheets integration for importing contact records in bulk.
- Developed a real-time CRM dialer accessible from contact records, integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented a CRM-wide soft-delete and Trash workflow with a 30-day retention period, replacing hard-delete behavior and using an AWS Lambda service to permanently remove records after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed a task cadence system that allows users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging (FCM) used for task notifications.
- Built account-level reporting dashboards and developed an MCP integration exposing CRM capabilities as tools, enabling the AI bot to retrieve CRM information and trigger supported operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur